

Hints of collapse - Part 1

The theory of tipping points

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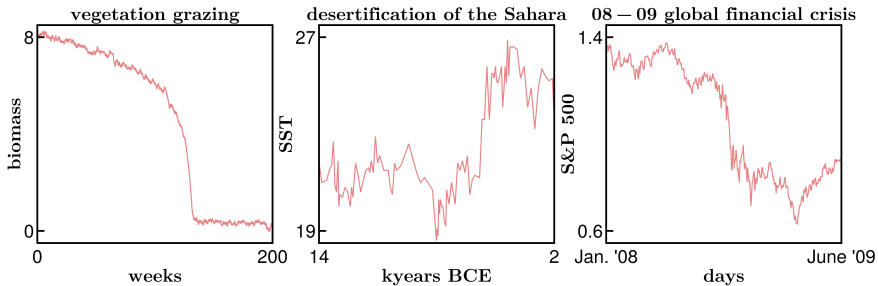
†Auckland Bioengineering Institute



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Collapsing systems



Main modelling tool

$x(t) \in \mathbb{R}^n$ is a state variable that evolves over time.

$\mu \in \mathbb{R}$ is a parameter.

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Initial value problem (IVP) made of a system of ordinary differential equations (ODEs) of the form

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The long-term behaviour of a solution is determined by the asymptotic limit of its evolution, i.e. $\lim_{t \rightarrow \infty} \phi(x_0, t)$.

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The emergent behaviour at population level is captured by a *revision protocol* named the **replicator equation**

$$\dot{x} = x(1 - x)(r_1(x) - r_2(x)) .$$

The perspective of the policymaker

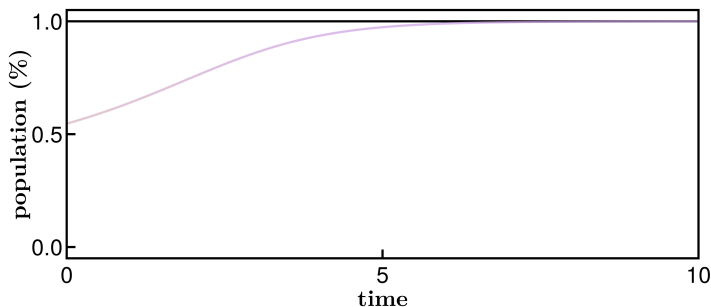
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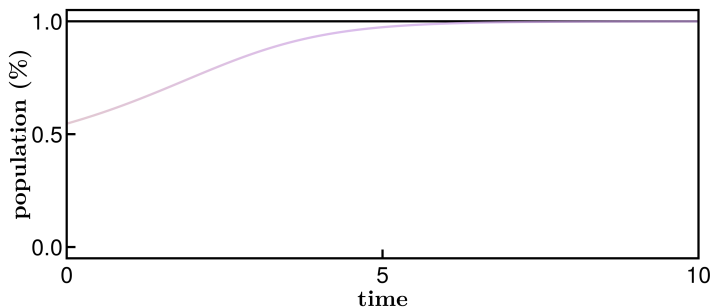


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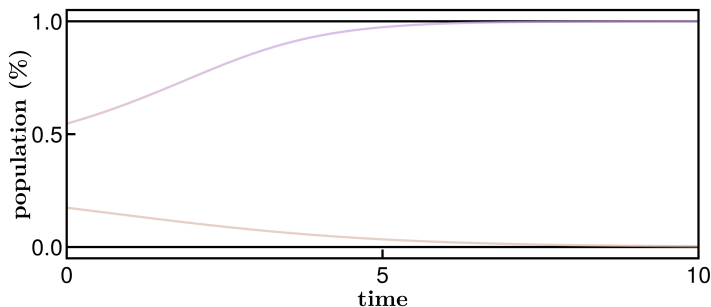


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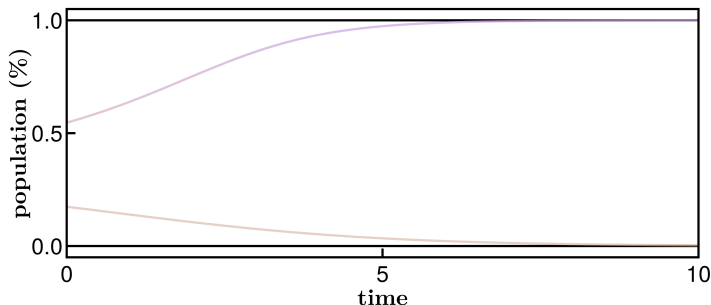
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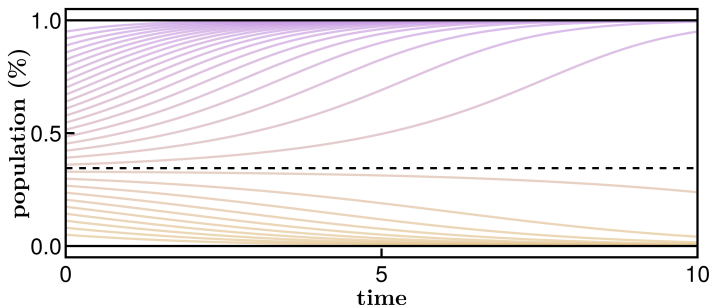
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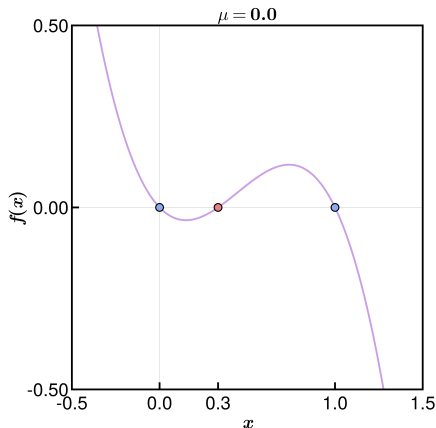
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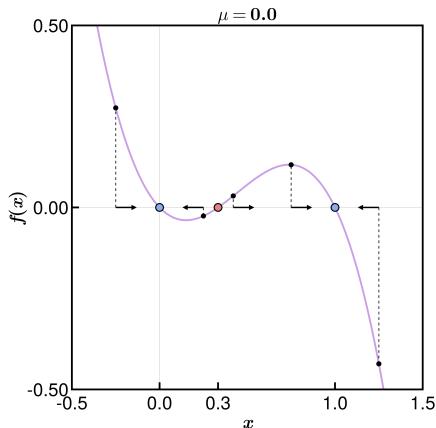
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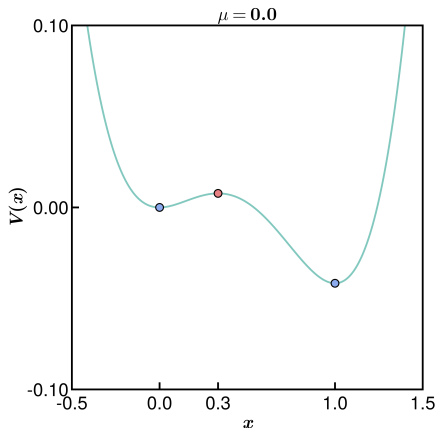
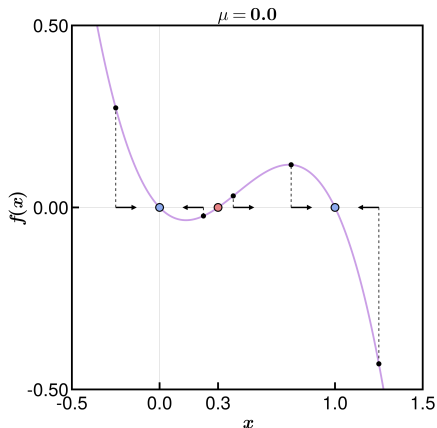
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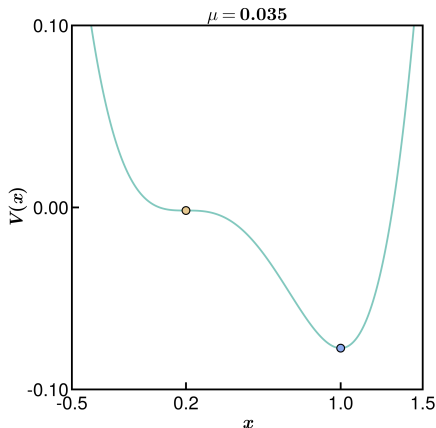
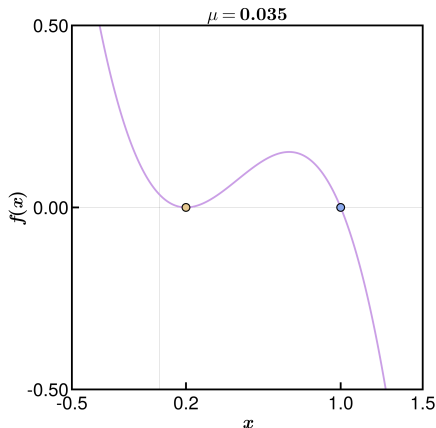
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Theorem n. 1 (existence and uniqueness)

Theorem (Picard-Lindelöf)

Let $\dot{x} = f(x, t)$ with $f : D \subseteq \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ continuous in t and Lipschitz in x and $x(t_0) = x_0$ being the initial datum. Then there exist a $\delta > 0$ s.t. the solution $\phi(x_0, t_0)$ exists and is unique on $\mathbb{B}_\delta(t_0)$.

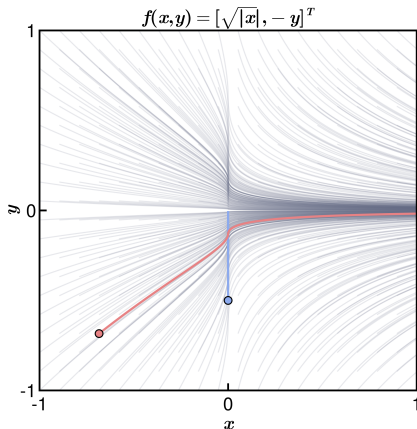
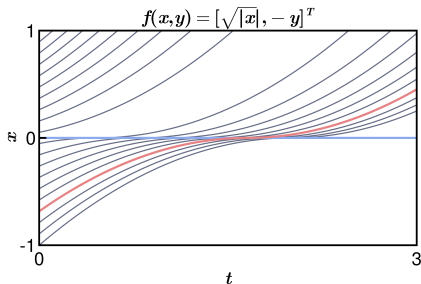
In other words, if there are two solutions ϕ_1 and ϕ_2 defined on the same time interval $\mathbb{B}_\delta(t_0)$ stemming out from the same initial condition x_0 , then $\phi_1 = \phi_2$.

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Theorem n. 2 (linear stability)

Theorem (Hartman-Grobman)

Let $\dot{x} = f(x; \mu)$ be a smooth dynamical system and x^* be a **hyperbolic** equilibrium of f . Then there exist a neighborhood $U \subset \mathbb{R}^n$ of x^* and a homeomorphism $h : U \rightarrow \mathbb{R}^n$ s.t. $h(\phi(x, t)) = \psi(h(x), t)$, where ψ is the solution stemming from the origin of the linearised system $\dot{x} = Df(x^*)(x - x^*)$.

The following vector field has three equilibria: $(0, 0)$, $(1, 0)$ and $(0, 1)$

$$\begin{cases} \dot{x} = x(1 - x - y), \\ \dot{y} = y(2 - x - 2y). \end{cases}$$

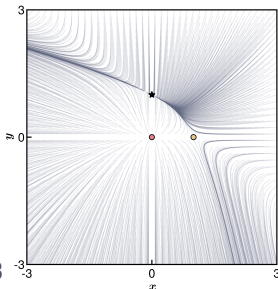
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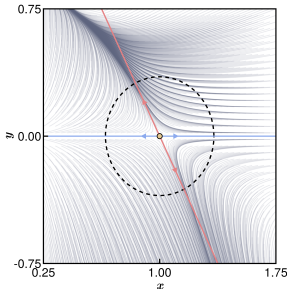
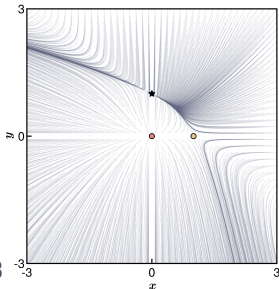
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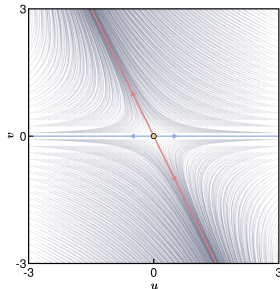
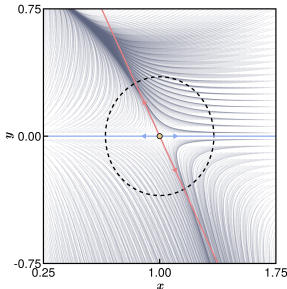
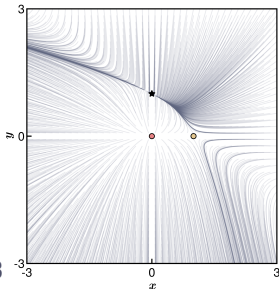
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Back to game theory

Let us rewrite

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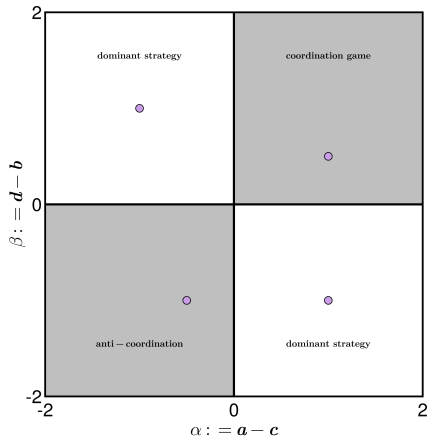
By choosing $(\alpha, \beta) \in \Gamma$ carefully we can steer our population to adopt the desired strategy (equilibrium selection).

Equilibrium selection

Values for (α, β) for which the phase flow qualitatively changes are locii of **transcritical bifurcations** .

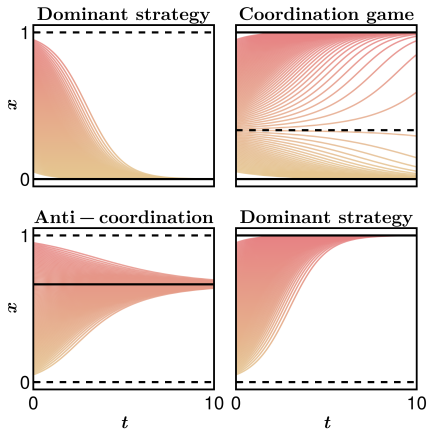
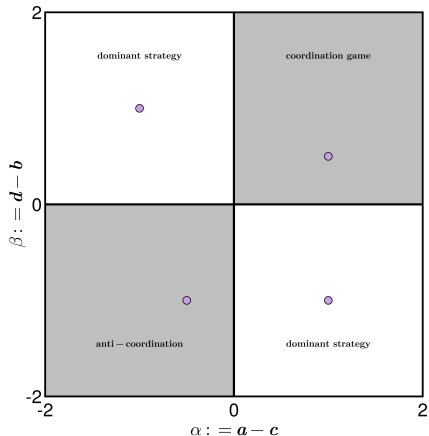
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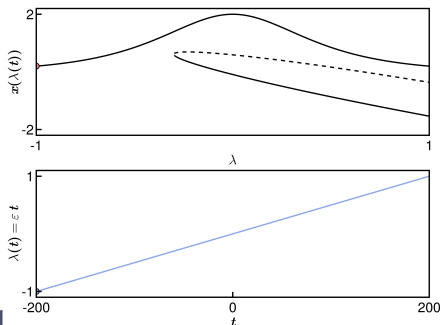
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$$f(x, \lambda) = -((x + a + b\lambda)^2 + c \tanh(\lambda - d)) \left(x - \frac{k}{\cosh(e\lambda)} \right),$$

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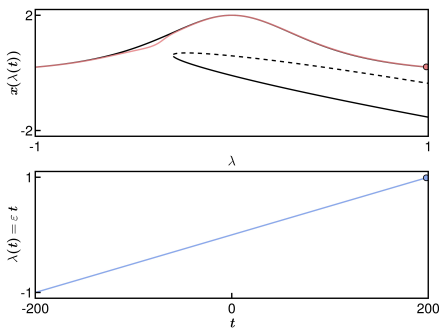
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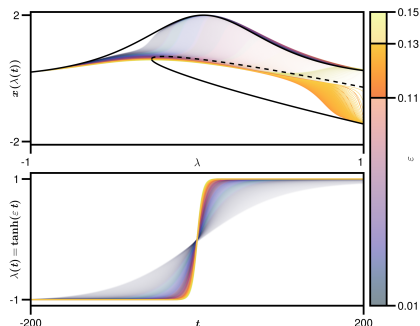
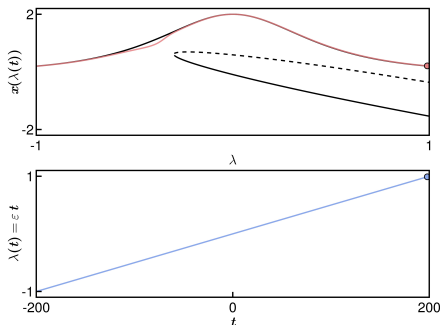
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Non-autonomous systems

We need to update (generalise) our notion of local attractors for non-autonomous systems, i.e. IVPs whose ODEs have an explicit time-dependence.

Definition (Pullback attractor)

Let $\dot{x} = f(x(t), t)$ be a non-autonomous system, then a family of sets $\{A(t)\}_{t \in \mathbb{R}}$ is a pullback attractor if:

- 1) each $A(t)$ is invariant under the cocycle: $\phi_{t,s}(A(s)) = A(t)$ for all $s \leq t$;
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A solution cocycle $\phi_{t,s}(x_0) = x(t)$ is defined so that, for all $s \leq t$, it satisfies

$$(\phi_{t,s} \circ \phi_{s,u})(x_0) = \phi_{t,s}(x_0), \quad u < s \leq t.$$

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Let $\dot{x} = f(x(t), t)$ be a non-autonomous system, then a family of sets $\{A(t)\}_{t \in \mathbb{R}}$ is a pullback attractor if:

- 1) each $A(t)$ is invariant under the cocycle: $\phi_{t,s}(A(s)) = A(t)$ for all $s \leq t$;
- 2) sets B in the past limit converge in forward time: $\lim_{s \rightarrow -\infty} |\phi_{t,s}(B) - A(t)| = 0$.

A solution cocycle $\phi_{t,s}(x_0) = x(t)$ is defined so that, for all $s \leq t$, it satisfies

$$(\phi_{t,s} \circ \phi_{s,u})(x_0) = \phi_{t,s}(x_0), \quad u < s \leq t.$$

Under mild conditions, there is a one-to-one relation between *classical* attractors $A \subset \mathbb{R}^n$ in the past limit autonomous system (i.e. when $t \rightarrow -\infty$) and pullback attractors $A(t)$ in the non-autonomous system.

Sketch: see whiteboard.

Theorem n. 3 (a sufficient condition)

Theorem (irreversible rate-induced tipping)

Consider a *frozen* vector field $f(x, \lambda)$ and a family of parameter shifts $\Lambda_\varepsilon(\lambda_-, \lambda_+)$. Suppose states $x \rightarrow x_\pm$, as $\lambda \rightarrow \lambda_\pm$, are connected by a path $(x(s), \lambda(s))$. If there are some μ s.t.

$$\overline{\{x(\mu) : \mu < \nu\}} \not\subset \{x_0 : |\phi(x_0, t) - x(\nu)| \rightarrow 0, t \rightarrow \infty\},$$

then there is a $\varepsilon_0 > 0$ s.t. $\lim_{t \rightarrow \infty} A(t) = x_+$ for $\varepsilon \in (0, \varepsilon_0)$ and $\lim_{t \rightarrow \infty} A(t) = y_+$ (with $y_+ \neq x_+$) for $\varepsilon \in [\varepsilon_0, +\infty)$.

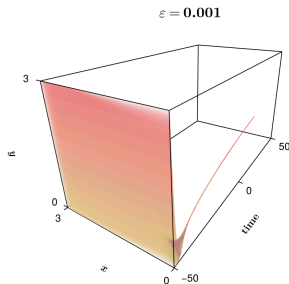
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Theorem n. 3 (a sufficient condition)

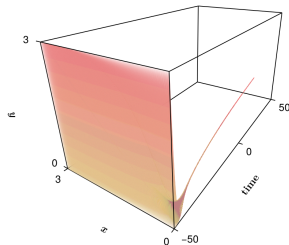
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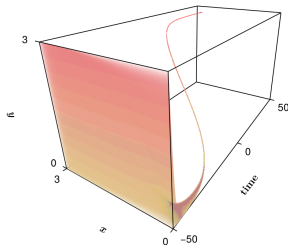
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$\varepsilon = 0.001$



$\varepsilon = 0.1$

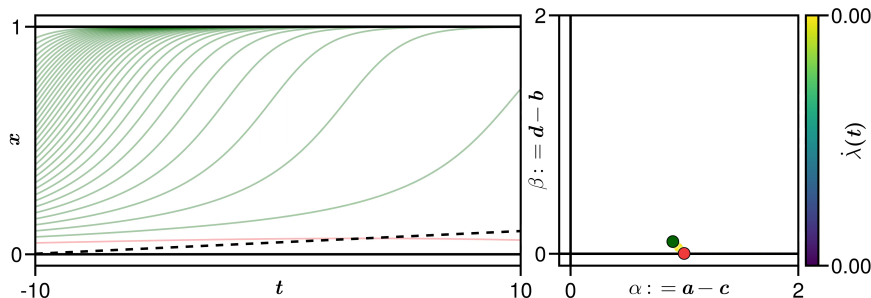


Forcing tipping as a policymaker

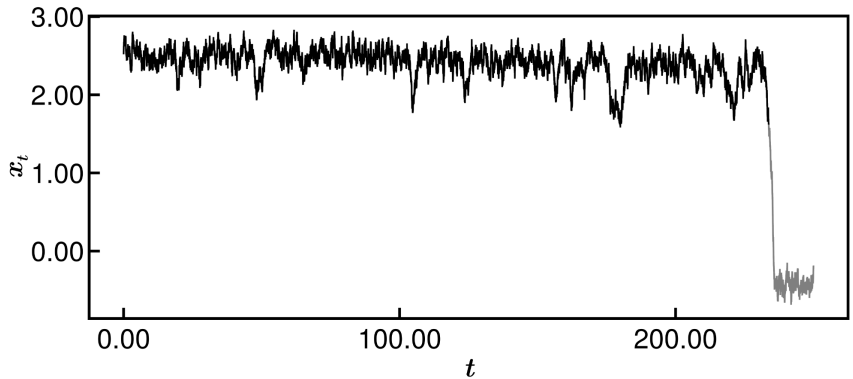
We further simplify our model by imposing $\alpha = 1 - \beta$ but at the same time we make it non-autonomous by letting $\beta = \lambda(t) = \frac{1}{2}(\tanh(\varepsilon t + \delta) + 1)$

$$\dot{x} = x(1-x)(x - \lambda(t)).$$

If we act fast enough (i.e. we pick ε large) we can make some trajectories to tip to the favorable strategy.



Where's the catch?



Stay tuned for Part 2!



C. Kuehn, Physica D **240** (2011).



L. Zino, et al., IEEE Tr. Aut. Control (2025).



P. Ashwin, et al., Nonlinearity **30** (2017).

Thank you.

► Go to the beginning

► Go to Theorem 1

► Go to Theorem 2

► Go to Theorem 3